

The Global Gender Gap in STEM Applications: Pipeline vs. Choice

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Abstract

Women make up only 35% of global STEM graduates, a share that has remained unchanged for a decade. To understand this disparity we separate gender differences in academic preparation (the pipeline) from gender differences in application decisions (the choice gap), a distinction that is rarely feasible because preparation, eligibility, and choice are typically intertwined. We compute the pipeline and choice gap across ten coordinated university admissions systems leveraging a key common institutional feature: in all of them eligibility is determined by academic performance, and assignment is implemented by a truth-telling mechanism. Focusing on high-achieving students, for whom access constraints are least likely to bind, allows us to isolate differences in application behavior in the absence of other objective barriers. Gender differences in the pipeline vary widely, ranging from female disadvantage in Uganda to female advantage in Sweden, and broadly track labor-market gender parity. By contrast, the choice gap is large, negative, and remarkably stable across settings that differ substantially in population size, economic development, and gender norms: even among top scorers, women are about 25 percentage points less likely than men to rank a STEM degree first. The result is similar when we consider listing a STEM program anywhere in the application portfolio rather than only first. Taken together, these findings suggest that women’s underrepresentation in STEM is shaped by factors that operate broadly across very different settings.

Keywords: gender inequality, STEM gender gap, centralized application platforms

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1. Introduction

Despite decades of progress in educational attainment, women remain substantially underrepresented in science, technology, engineering, and mathematics (STEM). In 2024, they accounted for only 35% of STEM graduates worldwide—a share that has barely moved in a decade (UNESCO, 2024). Explanations for this persistence are typically grouped into two channels: a *pipeline* channel—gender differences in academic preparation and access to selective STEM programs (Card and Payne, 2021, Aucejo and James, 2021, Humphries et al., 2023)—and a *choice* channel—gender differences in field preferences conditional on eligibility (Zafar, 2013).

Why focus on STEM in particular? Field-of-study choices have large implications for subsequent labor-market outcomes, and STEM-intensive trajectories are often associated with high private returns and strong demand for skills. Persistent gender gaps in entry into these fields are therefore relevant not only for equity, but also for efficiency, as losing talent in this area might lead to losses in innovation, productivity, and growth (Weinberger, 1999, Hoogendoorn et al., 2013, National Science Foundation, 2017, Hsieh et al., 2019).

Distinguishing pipeline from choice is empirically difficult, since it requires observing both program-specific eligibility and students’ ranked application decisions. While Delaney and Devereux (2019, 2021) demonstrate the value of using application data to study gender differences in college applications and STEM choices, existing research has largely been confined to single-country settings, limiting our ability to assess whether these patterns generalize across distinct institutional and cultural contexts.

This paper addresses that challenge by leveraging administrative microdata from centralized admissions systems in ten settings across five continents—Australia, Brazil, Chile, China, Finland, Greece, Spain, Sweden, Taiwan, and Uganda. Despite large differences in population size, economic development, and gender norms, these systems share a key institutional structure: students submit ranked preferences over college–major combinations through coordinated platforms (Neilson, 2024), and are assigned to the highest-ranked option for which they are eligible. Eligibility depends almost exclusively on academic per-

formance. This setting allows us to observe students’ ranked choices while comparing women and men with similar admission prospects.

Our empirical design focuses on high-achieving students whom we define following three alternative definitions: top 5%, top 10%, and top 20% of each setting’s academic performance distribution. We also examine two complementary measures of STEM choice, ranking a STEM program first, and ranking a STEM program anywhere in the preference list.

This distinction is important in the systems we study. Under deferred acceptance (DA), admissions outcomes depend on rank order as students are allocated to the highest-ranked option for which they are eligible. In standard DA environments with unconstrained lists, truthful ranking is optimal (Roth, 1982). In practice, however, list-length limits vary across countries. Brazil is an extreme case in which students can rank only two programs, while in Greece students can rank as many as they wish. Limited lists can generate strategic concerns, such as not listing the most preferred program first to insure admission. Several patterns in our data mitigate this concern. First, the qualitative results are similar across settings with very different list limits, including settings without binding limits. Second, our focus on top-performing students reduces strategic pressure because these applicants are more likely to qualify for a broad set of programs. Third, the choice gap appears both for first-choice and any-choice outcomes. Finally, we do not observe analogous patterns in other selective fields, such as business and law or health.

We begin by documenting substantial female underrepresentation among high-achieving STEM applicants across settings. We then decompose this pattern into a pipeline gap and a choice gap. We define the *pipeline gap* as the female-minus-male share among high-achieving students, and the *choice gap* as the female-minus-male difference in the probability of applying to STEM (first or anywhere in the list).

Pipeline patterns vary markedly across settings, and this conclusion is robust across the top 5%, top 10%, and top 20% definitions. In some countries the pipeline gap is negative, while in others it is positive. Yet women remain underrepresented among high-achieving STEM applicants even in settings where they constitute a larger share of top students than men. This implies that the

pipeline alone cannot account for the observed STEM underrepresentation.

By contrast, the choice gap is consistently large and negative. Regardless of whether we focus on first-choice STEM or any-choice STEM, high-achieving women are less likely than similarly talented men to rank STEM programs in every setting we study. In this sense, the central regularity in the data is not cross-country heterogeneity in the choice gap, but its persistence.

We also examine how pipeline and choice gaps correlate with cross-country gender parity in wages. We do this to assess whether the size of these gaps varies with gender norms, using labor-market wage parity as a proxy for norms in a domain that is likely to be particularly relevant for higher-education specialization decisions. We find that the pipeline gap tends to be large and negative in settings with low wage parity, and can turn positive in settings with high wage parity. By contrast, the choice gap remains large and negative across the parity distribution. These correlations are descriptive, but they reinforce the main empirical pattern. Differences in preparation vary strongly with context, whereas differences in STEM choices are much more stable. This stability in the choice gap suggests that important underlying drivers are not context-specific, but reflect elements present across very diverse settings.

Our contribution is to provide the first cross-national decomposition of the STEM gender gap into a pipeline gap (access and preparedness) and a choice gap (application decisions). This evidence bridges two strands of research. A first strand disentangles pipeline and choice within single settings (Delaney and Devereux, 2019, Card and Payne, 2021), but their narrow scope limits external validity. A second strand, typically in the form of international reports (OECD, 2017, Encinas-Martín and Cherian, 2023, UNESCO, 2024), documents STEM gender gaps across education systems but cannot separate pipeline from choice due to data limitations. By focusing on enrollment outcomes, these studies mix together pipeline, choices, and admissions. We assemble a novel global dataset that captures students’ intended fields of study rather than only their final enrollments. Centralized admissions systems provide ranked application lists, and because they operate variants of the deferred-acceptance algorithm, students have strong incentives to truthfully reveal their preferences. This allows us to observe what students want to study, not merely where they ultimately enroll.

Such a cross-country comparison has not been yet possible, as analyzing STEM choices directly requires harmonized application data—which has not been previously available at this scale. By harmonizing centralized admissions data from ten educational systems, we provide the first systematic cross-national decomposition of the STEM gender gap into pipeline and choice components. We show that while pipeline gaps vary considerably, the choice gap is strikingly stable, pointing to structural drivers that transcend local institutions and norms. These findings suggest that closing academic performance gaps, though important, will not by itself eliminate gender disparities in STEM.

2. Data

This section outlines the institutional context and data for the ten settings we study: Australia, Brazil, Chile, China, Finland, Greece, Spain, Sweden, Taiwan, and Uganda. In Online Appendix ?? we provide additional details on each setting. It is worth noting that while in most settings our data covers the entire country, in Australia and China we only have access to data covering specific regions. In Australia, we have data from Victoria, a state that accounts for 25% of the country’s population and serves as a strong proxy for the nation as a whole. In China, we have data from Ningxia, an autonomous region containing just 0.5% of the national population, representing a more rural, less economically developed territory than the national average.

More generally, the scope of institutional coverage is broad in the systems we study. In many settings, all universities participate in the centralized admission system; when coverage is partial, the participating institutions are typically the more selective ones. For example, in Brazil all public universities participate, while in Chile all public universities and the selective private universities participate. Overall, our data cover the institutions that high-achieving students are most likely to attend. For additional institutional details, see Online Appendix A.

Panel A of Table 1 shows sharp contrasts across these countries in size, income, human development, inequality, and gender norms. China is by far the largest country in our sample (1.4 billion people), followed by Brazil (209

million), Spain (47 million), and Uganda (40 million). Sweden (10 million) and Finland (5.5 million) are the smallest.

Because the data available for each setting cover different periods, in Online Appendix [Appendix D](#) we report estimates for the cohort closest to 2018 in each country. The results remain very similar to our baseline estimates.

Australia, Taiwan, Finland, and Sweden are among the wealthiest countries in the world, with GDP per capita between USD 55,000 and 65,000. They also score very high on the United Nations Human Development Index (HDI), which ranges from 0 to 1, with values of 0.8 or above classified as “very high”; all four exceed 0.92. Uganda, by contrast, has a GDP per capita of USD 3,500 and an HDI of 0.53, placing it in the “low development” category. Brazil, Chile, China, and Greece fall in between, with GDP per capita between USD 20,000 and 41,000 and HDI values between 0.75 and 0.88. Inequality also varies widely across the sample. Brazil, Chile, and Uganda have Gini indexes above 0.42, while Australia, Greece, Finland, Spain, and Sweden are below 0.35.

To characterize countries according to gender norms that could be relevant for our study, we built a gender wage parity index, defined as 1 minus the gender wage gap. As shown in Panel A of Table [1](#), this index ranges from 0.517 in Uganda to 0.965 in Greece. The construction and sources of this index are described in detail in Section [4.4](#).

Despite cross-country differences, the admission systems we study share two features. First, they allocate most university seats through centralized platforms using variants of the deferred acceptance algorithm. This means students submit ranked lists of preferred programs and are placed in the highest option for which they qualify. Importantly, eligibility is almost exclusively based on academic performance, typically high school grades and admission exam scores. Our data, drawn from the admissions agencies, include both students’ ranked applications and their performance records.

Panel B of Table [1](#) summarizes key differences in institutional coverage and pricing. In some settings, centralized admissions cover all universities in the system—for example, Australia 21/21, Greece 41/41, Sweden 41/41, Taiwan 67/67, and Uganda 8/8—while in others coverage is partial—for example, Brazil

132/2448, Chile 34/60, China 1,252/2,663, and Spain 50/86.¹ Tuition policies also differ sharply. Finland, Greece, Sweden, and Brazil report no tuition fees among the universities participating in the centralized admission system, whereas Australia, Chile, Spain, China, Taiwan, and Uganda report tuition charges. Different forms of financial aid are available in all settings.

Finally, Panel C of Table 1 describes the centralized admission systems we study. In all settings, students apply to specific college-major combinations, typically from hundreds of options—ranging from 609 in Greece to more than 18,000 in China. The number of programs students may rank also varies. In Brazil they can list only two, while in Greece there is no limit. All of these systems are based on deferred acceptance (DA) mechanisms, which provide students with incentives to report their preferences truthfully, allowing us to recover their underlying rankings of programs.²

Panel C also shows that the admission systems in our data vary in scale. The counts we present there correspond to yearly averages. Brazil is the largest system in our sample with 2.71 million applicants per year, followed by Spain with 171 thousand, and Chile with 132 thousand. At the other extreme, Uganda with 18.1 thousand and the Victoria region in Australia with 38.8 thousand have the smallest annual applicant flows. Relative to the annual pool of university applicants, admitted STEM students represent between 2.4% (Brazil) and 38.4% (Taiwan).³

Women are generally more likely than men to apply. Except for Taiwan (49.8%) and Uganda (43.9%), female applicants outnumber males, reaching 60.1% in Finland and 57.4% in Sweden. These patterns mirror findings for

¹None of Brazil’s 2,152 private higher education institutions use centralized admissions.

²While DA is strategy-proof when students can rank all programs, this property breaks down if the preference list is restricted (Fack et al., 2019). In practice, these constraints are rarely binding. In all systems except Brazil, fewer than 5% of students exhaust their lists. Brazil is the main exception, as the SISU system allows only two programs per round. However, SISU operates through an iterative version of DA, in which applicants may resubmit choices over multiple rounds. Both theoretical and experimental evidence show that this iterative structure sustains truthful reporting among feasible options and delivers stable outcomes (Bó and Hakimov, 2019, 2022). Thus, reported choices can be interpreted as students’ preferred options among the set of programs they could plausibly attend. For further discussion, see Barahona et al. (2023).

³Note that in Brazil only public universities participate of the centralized admission system. The 2.4% share of students admitted to STEM programs ignores students admitted in STEM degrees offered by private universities.

the United States (Goldin and Katz, 2008).

Table 1: Institutional Characteristics

	Australia (1)	Brazil (2)	Chile (3)	China (4)	Finland (5)	Greece (6)	Spain (7)	Sweden (8)	Taiwan (9)	Uganda (10)
Panel A										
Setting Characteristics										
Population (millions)	24.6	209.5	18.7	1,402.8	5.5	10.7	46.8	10.2	24.0	40.1
GDP per capita (thousands)	56	21	31	24	56	41	50	62	66	4
Human Development Index	0.937	0.764	0.849	0.755	0.937	0.881	0.905	0.943	0.925	0.534
Gini index	33.7	53.9	44.4	38.5	27.3	32.9	34.7	30.0	34.2	42.8
Wage parity index (1 – wage gap)	0.885	0.900	0.837	0.786	0.832	0.965	0.941	0.901	0.671	0.517
Panel B										
University System										
N of Institutions	21/21	132/2448	34/60	1,252/2,663	36/38	41/41	50/86	41/41	67/67	8/8
Tuition fees	Yes	No	Yes	Yes	No	No	Yes	No	Yes	Yes
Financial aid	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Panel C										
Admission System										
Options available (yearly avg.)	1,078	6,310	1,423	18,671	1,458	609	2,169	15,374	1330	149
Max. apps	12	2	10	90	6	No limit	12	20	100	6
N of applicants in a year (avg.)	38.8	2,711.0	132.2	52.4	69.6	68.0	171.6	108.2	72.1	18.1
N of admitted students in STEM (avg./year)	7.9	65.0	30.7	18.9	8.1	17.9	32.6	13.6	27.7	0.7
Female share (apps)	55.8%	57.1%	53.0%	55.5%	60.1%	55.8%	56.8%	57.4%	49.8%	43.9%
Data coverage	2009/10	2016	2012/18	2018	2016/20	2003/12	2020/24	2017	2010	2021/22

Notes: The table provides summary statistics characterizing the settings in our sample and their university admission systems. Panel A provides general information on each setting, panel B characterizes their university systems, and panel C describes their university admission systems. Population, GDP per capita, HDI, and Gini come from World Economics (<https://www.worldbank.org/GDP-Per-Capita>), United Nations Development Programme (<https://hdr.undp.org/data-center>), and the World Economic Forum (https://www3.weforum.org/docs/WEF_GGGR_2023.pdf). The wage parity index is defined as 1 minus the gender wage gap; data sources and construction details are reported in Section 4.4. Australia corresponds to Victoria and China corresponds to Ningxia Province. The row “N of Institutions” reports institutions covered by the centralized system over the total number of institutions in each setting, highlighting differences in coverage across countries. Numbers of applicants and admitted students are in thousands and represent yearly averages.

There are other relevant institutional differences in exam design, score scales, or tested content that might contribute to the gender differences we study. However, understanding how these differences contribute to pipeline or choice gaps is beyond the scope of this paper. Our aim is instead to document within-system gender differences and to understand whether, conditional on having similar admission probabilities, there is a persistent gap in STEM applications. For additional institutional details, see Online Appendix ??.

3. Empirical strategy

This paper examines gender differences in representation among STEM applicants across ten settings that considerably differ in population size, economic development, inequality, and gender norms. A key feature that all these settings share is the use of centralized university admission systems, where admissions depend on the ranked list of preferences that students submit and on their academic performance. This institutional structure means that students with similar academic performance face similar admission probabilities.

Leveraging these features, we focus on high-achieving students, defined as those in the top 20%, 10%, and 5% of their cohort based on the mandatory components of each setting’s academic performance measure (e.g., university

admission exam, high school exit exam, high school GPA). This approach allows us to include all applicants, regardless of their intended field. We focus on top students because, conditionally on applying, these students are most likely to gain admission to and succeed in selective STEM programs, which are associated with large economic and social returns. Academic performance deciles are computed within each cohort and centralized admission system.

In this paper, we define programs as STEM based on ISCED-2013 two-digit fields of education and training⁴, grouping programs in natural sciences, mathematics and statistics (code 05), information and communication technologies (code 06), and engineering, manufacturing and construction (code 07) under this category. In Online Appendix [Appendix C](#), we provide additional results focusing on business and law (code 04) and on health (code 09).

Our analysis begins by characterizing the gender composition of high-achieving STEM applicants. We then decompose these gender differences by examining two gaps:

1. The pipeline gap: difference between women’s and men’s representation among students in the top 5%, 10%, and 20% of each setting’s academic performance distribution.
2. The choice gap: difference in the share of high-achieving women and men who rank a STEM program either as their top choice or at any rank of their preference list.

We conclude by examining whether these gaps correlate with cross-country wage parity, using a labor-market measure that is directly relevant for higher-education specialization decisions.

⁴See UNESCO Institute for Statistics, *International Standard Classification of Education (ISCED)*: <https://uis.unesco.org/en/topic/international-standard-classification-education-isced>.

4. Results

4.1. Female representation among top STEM applicants

Figure 1 reports the share of female students among high-achieving STEM applicants. We define high-achieving students as those in the top 5%, 10%, and 20% of the academic performance distribution in each system, and we consider two complementary STEM applicant definitions: (i) ranking a STEM program first, and (ii) ranking a STEM program anywhere on the application list. This distinction is meaningful in our settings because admissions are based on deferred acceptance, which assigns students to the highest-ranked program for which they are eligible. As a result, ranking STEM first is especially informative about the intensity of STEM preferences and has a more direct implication for admissions outcomes. By contrast, the any-choice measure captures broader STEM interest and helps distinguish students who do not apply to STEM at all from those who include STEM but rank it lower.

Across systems, women are generally underrepresented among high-achieving STEM applicants under both definitions, although the magnitude of the gap varies substantially across contexts.

Focusing on students who rank STEM first, underrepresentation is clear. In five of the ten settings, women account for less than 35% of high-achieving STEM applicants at each performance threshold (top 5%, 10%, and 20%). Chile exhibits the lowest female representation, with women comprising roughly one-quarter of high-achieving STEM applicants. In the remaining settings, female underrepresentation is still there but is smaller. The only exception is Greece, where women represent 53% of top-5% students ranking STEM first, though this share falls below 50% at the top-10% and top-20% thresholds.

When we broaden the outcome to any-choice STEM applicants, the main pattern remains. Women are still underrepresented among high-achieving STEM applicants in most settings. Female shares are typically somewhat higher than under the first-choice measure. The largest shifts appear in China and Taiwan. In China, female representation rises by 13 to 16 percentage points under the any-choice definition, yet remains below 50%. Similarly, in Taiwan, it rises by 5 to 8 percentage points and also stays below 50%. Greece is the only case where

women are consistently the majority under this measure, with shares between 51% and 56% across performance thresholds.

Decomposing the sources of female underrepresentation. The female shares illustrated in Figure 1 represent the combined effect of two distinct forces. First, the representation of women among high-achieving students overall (the pipeline). Second, the propensity of high-achieving women to rank STEM programs relative to high-achieving men (the choice gap). The observed female shares in STEM result from the interaction of these two mechanisms: even if women are well-represented among high-achieving students, they may still be underrepresented in STEM if they choose STEM less frequently. Conversely, underrepresentation can persist even if the pipeline is balanced, if the choice gap is sufficiently large. To understand which force drives the observed patterns across our diverse settings, we now decompose female underrepresentation by analyzing each factor independently.

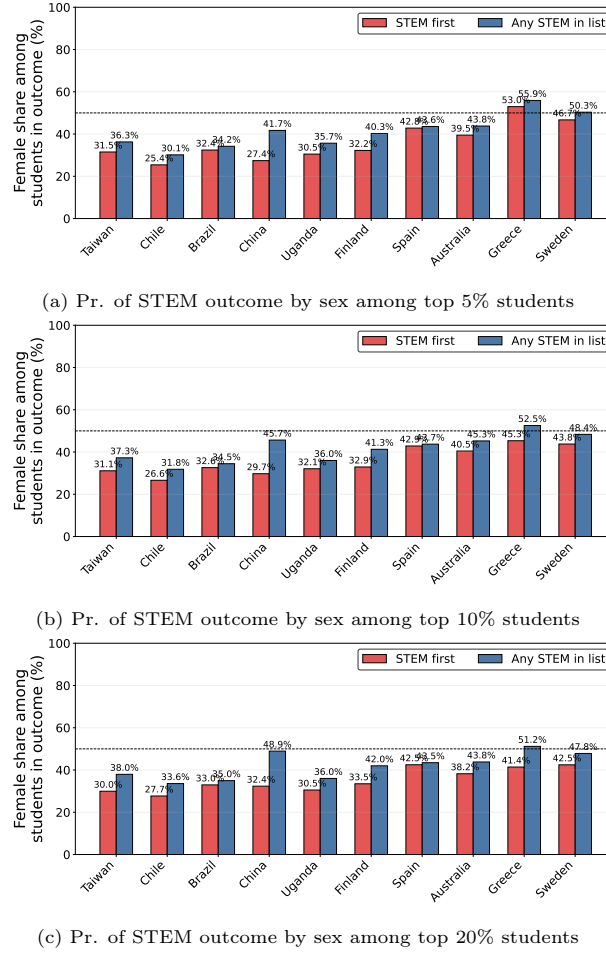
What drives these gender disparities and their variation across settings? Female underrepresentation among high-achieving STEM applicants could stem from two distinct sources: the *pipeline gap*—women being underrepresented among the high-scoring students who likely qualify for selective programs—or the *choice gap*—high-achieving women being less likely than their male counterparts to select STEM fields when applying to university. To disentangle these mechanisms, we next analyze each gap separately across our diverse settings.

4.2. The pipeline gap

Figure 2 illustrates the *pipeline gap*. Panels A, C, and E report female and male shares among students in the top 5%, 10%, and 20% of the academic performance distribution, respectively (blue bars for men and yellow bars for women). Panels B, D, and F report the corresponding pipeline gaps—defined as the difference between female and male shares—for each performance threshold.

In only three of the ten systems we study—Chile, Taiwan, and Uganda—women are underrepresented among high-achieving students. In Australia, Brazil, China, and Finland, female and male shares are approximately balanced, while in Greece, Spain, and Sweden women are overrepresented. The largest negative pipeline gap is observed in Uganda, where the female share

Figure 1: Gender Shares among STEM applicants at the Top of the Academic Performance Distribution



Notes: In each panel, bars report the female share among students observed in each outcome group (STEM first choice and any STEM in the ranked list), separately by country. The sample in panel (a) is students in the top 5% of each country's academic performance distribution, panel (b) uses the top 10%, and panel (c) uses the top 20%. Percent values above bars report the exact female share for each country-outcome cell. Outcomes are computed from harmonized administrative admissions counts using country-specific preferred sample variants (Australia: reply_docx; Brazil: reply_tables; Chile: chile_2012_2018; China: china_ningxia_2018; Finland: main_pooled; Greece: years_2003_2012; Sweden: reply_docx_2017; Taiwan: reply_pdf_2010; Uganda: reply_docx). Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan (N=72,067), Chile (N=1,747,925), Brazil (N=2,711,049), China (Ningxia) (N=52,427), Uganda (N=36,048), Finland (N=347,904), Australia (N=77,648), Greece (N=680,035), and Sweden (N=108,237). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

among high-achieving students is 20 to 25 percentage points below the male share across the top 5%, 10%, and 20% groups. The largest positive pipeline gap is observed in Sweden, where the female share is roughly 30 percentage points above the male share.

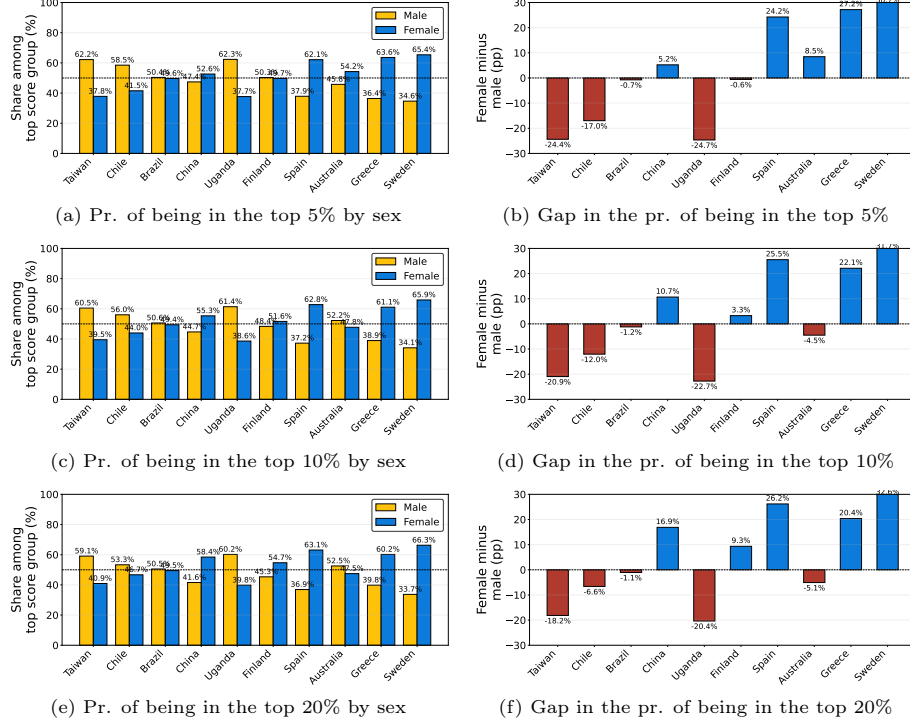
When comparing Figures 1 and 2, it is clear that the *pipeline gap* cannot fully explain differences in gender representation among STEM applicants. Even in settings where women outnumber men among high-achieving students, female representation among STEM applicants remains lower than male representation. More broadly, in every setting, women’s share among high-achieving STEM applicants—whether measured using first-choice or any-choice STEM preferences—is below their share among high-achieving students overall. This pattern indicates that factors beyond academic performance are influencing gender disparities in STEM applications.

4.3. *The choice gap*

Following the analytical framework established by Delaney and Devereux (2019), we examine the *choice gap*—the extent to which high-achieving women and men differ in their likelihood of ranking STEM programs first or including them anywhere on their application lists. Figures 3 and 4 illustrate these patterns for first-choice STEM and any-choice STEM outcomes, respectively. In each figure, panels A, C, and E report female and male shares among students in the top 5%, 10%, and 20% of the academic performance distribution, while panels B, D, and F report the corresponding female-minus-male choice gaps.

Focusing on first-choice STEM applications is particularly informative because, in the settings we study, the position at which students rank a program matters. The deferred acceptance algorithms used in these systems assign students to the highest-ranked program for which they are eligible. As a result, ranking a STEM program first is especially informative about the intensity of STEM preferences and has a more direct implication for admissions outcomes. By contrast, the any-choice measure captures broader STEM interest and helps distinguish students who do not apply to STEM at all from those who include STEM but rank it lower.

Figure 2: Gender Shares among Top Students and the Pipeline Gap



Notes: Panels (a), (c), and (e) report male and female shares among students in the corresponding high-performance group in each country (top 5%, top 10%, and top 20%, respectively). Panels (b), (d), and (f) report the pipeline gap, defined as female share minus male share in the same group. Positive gap values indicate female overrepresentation among top performers; negative values indicate female underrepresentation. Percentage labels on bars report exact values; in gap panels labels are placed above positive bars and below negative bars. Samples and harmonization follow the same preferred country variants described in Figure 1. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan ($N=72,067$), Chile ($N=1,747,925$), Brazil ($N=2,711,049$), China (Ningxia) ($N=52,427$), Uganda ($N=36,048$), Finland ($N=347,904$), Australia ($N=77,648$), Greece ($N=680,035$), and Sweden ($N=108,237$). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

Deferred acceptance algorithms are strategy-proof when students face no limit on the number of applications they can submit (Roth, 1982). In most of our settings, however, students do face application limits. In Brazil, for instance, students can submit only two applications, whereas in Greece applicants can submit as many as they wish. Because eligibility in all systems depends on academic performance, our focus on high-achieving students is useful under these constraints. High-achieving students are less likely to be affected by strategic concerns, as they are more likely to be eligible for most of the programs to which they apply. However, in Online Appendix [Appendix B](#), we show that in most settings the choice gap persists across the full performance distribution.

In contrast to the substantial cross-country differences observed for the *pipeline gap*, the *choice gap* is negative in every setting we study. Across all systems, high-achieving female students are less likely than their male counterparts both to rank a STEM program first and to include STEM anywhere on their application lists. Focusing on first-choice rankings, the gap exceeds 20 percentage points in eight of the ten educational systems in our sample. At the extremes, Uganda exhibits the smallest gap (11 to 16 percentage points), whereas China exhibits the largest (32 to 38 percentage points).

A similar pattern emerges when we measure the *choice gap* using STEM applications in any position of the preference list. Although the gap is smaller than under the first-choice measure, it remains substantial. In seven of the ten educational systems we study, the gap still exceeds 20 percentage points. In the remaining systems, it is smaller but nonetheless sizable. Taiwan and Uganda are the settings where these gender differences are the smallest, with them varying between 8 and 14 percentage points.

The differences we document in this section are specific to STEM degrees. In Online Appendix [Appendix C](#), we show that when looking at other selective fields associated with high-paying careers, such as business and law, there is no systematic choice gap—independently on whether we measure it by first preferences or for preferences listed in any position of the application list. In addition, when looking at health degrees, we find that the pattern is reversed. High-achieving women are systematically more likely than high-achieving men to rank health programs first or anywhere in their application list.

The striking consistency of the negative *choice gap* across settings that differ in size, economic development, and cultural context raises an important question: to what extent do broader societal factors, such as gender norms, explain the variation we observe in both the pipeline and choice gaps? We explore this question next by examining the relationship between these gaps and a measure of gender parity.⁵

4.4. The pipeline gap, the choice gap, and gender wage parity

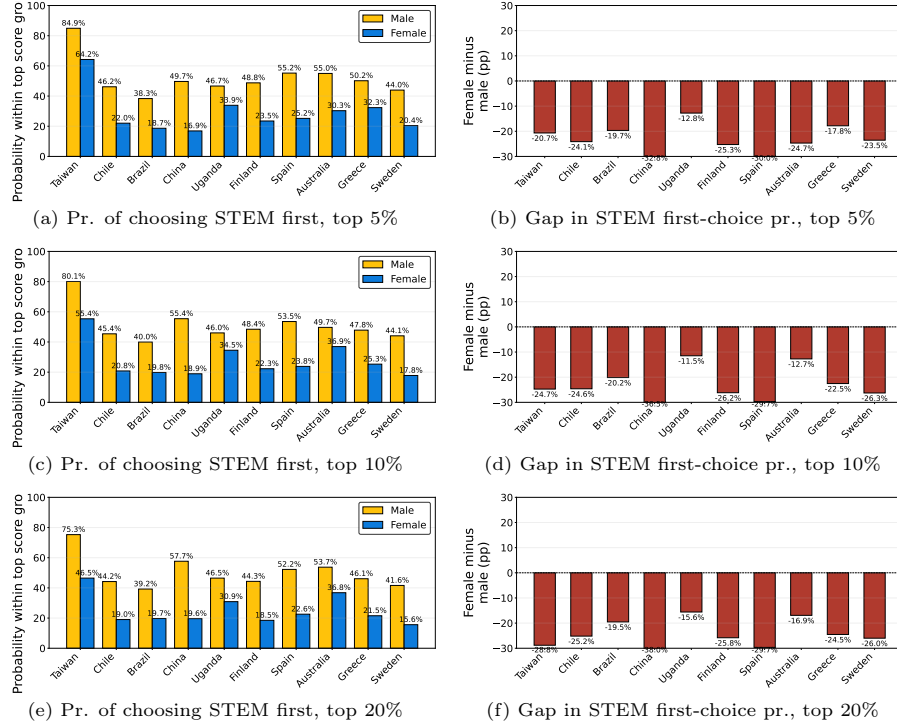
Gender norms and labor-market conditions are often cited as potential drivers of gender differences in educational outcomes (Akerlof and Kranton, 2000, Guiso et al., 2008, Bertrand, 2020). To examine this relationship in our data, we study correlations between the *pipeline* and *choice gaps* and a cross-country wage parity index. To construct this index, we collected the most recent estimate of the total gender wage gap reported by the International Labor Organization (International Labour Organization, 2026). This information was available for all settings in our sample except Australia, China, and Taiwan. For Australia and Taiwan, we rely on data reported on official websites (Workplace Gender Equality Agency, 2026, Directorate-General of Budget, Accounting and Statistics (DGBAS), Taiwan, 2016), while for China we use the estimate reported in Zhang and Dong (2025). The index used in the figure is one minus the gender wage gap, so values closer to one imply greater parity. Figure 5 illustrates these relationships.

Panel A shows a positive relationship between the female share among students in the top 10% of the academic performance distribution and wage parity. In many of the countries with higher wage parity, female shares among high-achieving students exceed male shares. As a result, the *pipeline gap* narrows and can even turn positive as wage parity increases.

In contrast, panel B shows that the relationship between wage parity and

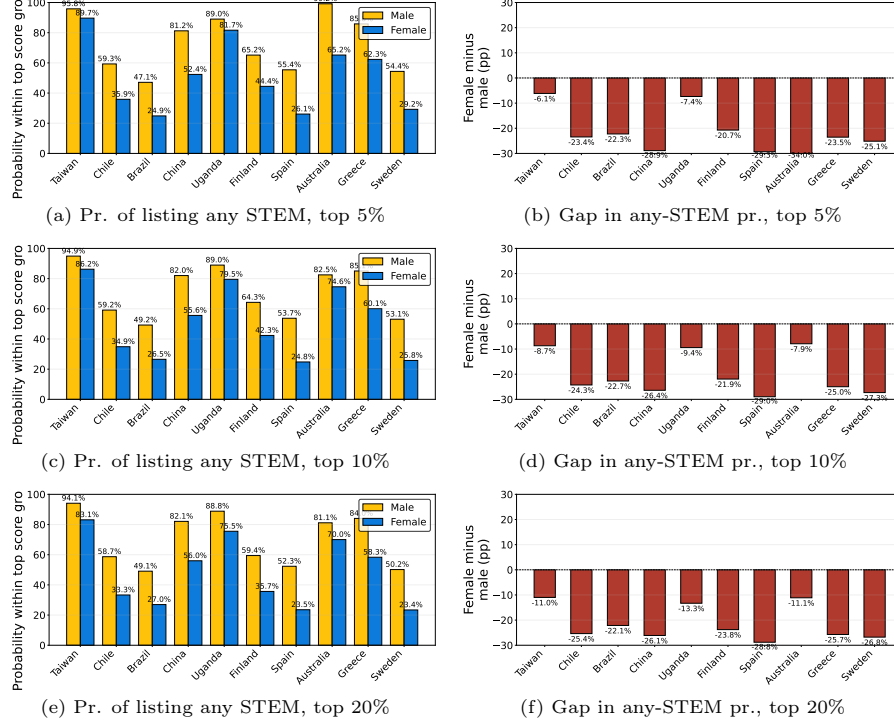
⁵This section uses application data covering different time periods. While in some setting we have data from the late 1990s and early 2000s, in others the data covers more recent years. In Online Appendix [Appendix D](#) we present results focusing on the year closest to 2018. Not all countries have data for that year, but the results remain similar when we focus on that year rather than in the whole sample. This suggests that the choice gap has not varied substantially in recent decades.

Figure 3: STEM first-choice probabilities and gender gap across the academic performance distribution



Notes: Panels (a), (c), and (e) show, by country, the probability that applicants rank a STEM program first in their preference list, separately for male and female students within the top 5%, top 10%, and top 20% of the academic performance distribution. Panels (b), (d), and (f) show the corresponding gender gap defined as female probability minus male probability (percentage points). Negative values indicate lower STEM first-choice propensity among women. For Chile, probabilities are conditioned on students with at least one submitted college application (while score groups are still defined on the full score-holder distribution). Percentage labels on bars report exact values; in gap panels labels are above positive bars and below negative bars. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan (N=72,067), Chile (N=1,747,925), Brazil (N=2,711,049), China (Ningxia) (N=52,427), Uganda (N=36,048), Finland (N=347,904), Australia (N=77,648), Greece (N=680,035), and Sweden (N=108,237). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

Figure 4: Any-STEM application probabilities and gender gap across the academic performance distribution

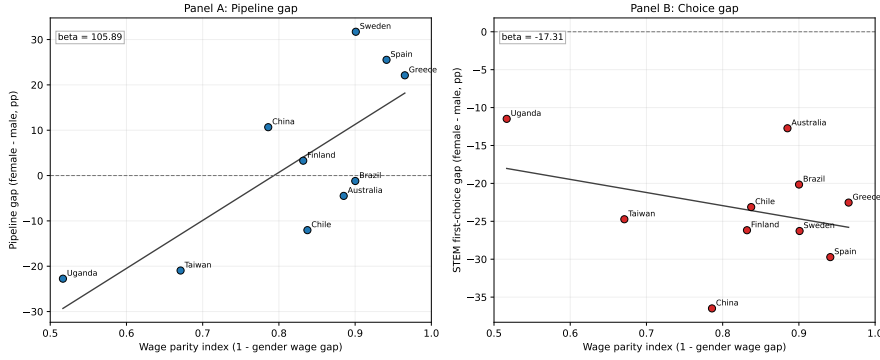


Notes: Panels (a), (c), and (e) report, for each country, the probability that an applicant includes at least one STEM program anywhere in the rank-order preference list, by sex and performance group (top 5%, top 10%, and top 20%). Panels (b), (d), and (f) report the female minus male difference in that probability (percentage points). Values are normalized by the number of applicants of each sex in the corresponding top-performance group. For Chile, probabilities are conditioned on students with at least one submitted college application (while score groups are still defined on the full score-holder distribution). Percentage labels on bars report exact values; in gap panels labels are above positive bars and below negative bars. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan ($N=72,067$), Chile ($N=1,747,925$), Brazil ($N=2,711,049$), China (Ningxia) ($N=52,427$), Uganda ($N=36,048$), Finland ($N=347,904$), Australia ($N=77,648$), Greece ($N=680,035$), and Sweden ($N=108,237$). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

the *choice gap* is weak. The choice gap remains negative and sizable across all systems, even where wage parity is relatively high. Moreover, the estimated slope, although small, is if anything negative.

Taken together, these patterns indicate that wage parity—which is closely associated with prevailing gender norms—is strongly correlated with the pipeline gap, but not with the choice gap. This implies that even in relatively progressive settings, high-achieving women remain considerably less likely than similarly talented men to apply to STEM degrees. Although the evidence in this paper is descriptive, the persistence of this choice gap across such diverse contexts suggests that an important part of female underrepresentation in STEM is driven by factors that are common across settings. In the next section, we discuss potential mechanisms that future research could examine more directly.

Figure 5: Pipeline and Choice Gaps vs Wage Parity Index



Notes: Panel A plots the relationship between the pipeline gap and a wage parity index. The pipeline gap is defined as the female share minus the male share among students in the top 10% of the academic performance distribution. Panel B plots the relationship between the choice gap and the same wage parity index used in panel A. The choice gap shown here corresponds to the difference in the share of high-achieving female and male students ranking a STEM degree at the top of their application list. The wage parity index on the x-axis is computed as 1 minus the gender wage gap; values closer to 1 therefore indicate higher wage parity. Wage gap figures come from ILO (latest available year by country) ([International Labour Organization, 2026](#)), complemented with country-specific official sources for Australia and Taiwan ([Workplace Gender Equality Agency, 2026](#), [Directorate-General of Budget, Accounting and Statistics \(DGBAS\), Taiwan, 2016](#)) and a published estimate for China ([Zhang and Dong, 2025](#)).

4.5. Potential mechanisms behind the choice gap

The remarkable stability of the choice gap across diverse contexts suggests that the underlying mechanisms are not context-specific. While our data cannot directly test which mechanisms drive this pattern, several factors could

contribute to this persistent gap. First, information asymmetries may play a role: women may have less access to information about STEM careers, salaries, and job characteristics, or may receive different career guidance (Ahn et al., 2019, Exley et al., 2024, Hastings et al., 2016). Second, gender stereotypes and lack of female role models in STEM fields may affect women’s self-efficacy and sense of belonging, even among high-achieving students (Carlana, 2019, Reuben et al., 2014). Third, women may anticipate discrimination in STEM workplaces or expect different treatment than men (Lepage et al., 2024, Lavy and Megalokonomou, 2024). Finally, women may systematically value different job attributes—such as flexibility, stability, and work-life balance—than men (Zafar, 2013, Wiswall and Zafar, 2018).

Distinguishing between these mechanisms is an important area for future research. Our centralized admissions data could potentially be enriched with survey information on students’ career expectations, self-efficacy beliefs, and preferences for job attributes, following the approach of (Zafar, 2013). Such data would allow researchers to test whether the choice gap reflects differences in information, expectations, or fundamental preferences. Understanding which of these mechanisms drive the choice gap is crucial for designing effective interventions.

5. Discussion and conclusion

The gender disparities we document in university applications matter for both equity and efficiency. Because returns to higher education vary by field, women’s underrepresentation in STEM—where returns are especially high—likely sustains labor market gaps. From an efficiency perspective, improving the gender balance in applications across fields could lead to a better allocation of talent and ultimately boost economic growth. Attracting more women into high-skill fields where they have been historically underrepresented could therefore yield substantial gains in productivity and aggregate output (Hsieh et al., 2019, National Science Foundation, 2017, Weinberger, 1999, Hoogendoorn et al., 2013).

Our main contribution is to decompose women’s underrepresentation in

STEM into a pipeline gap and a choice gap. To study the pipeline gap, we use alternative definitions of high-achieving students—top 5%, 10%, and 20%—and show that substantial cross-country variation in female representation among top students appears under all definitions. For example, under the top-10% definition, women represent less than 40% of high-achieving students in Uganda but more than 60% in Sweden. The pipeline gap alone therefore cannot explain women’s underrepresentation in STEM. We then study the choice gap by comparing the probability that high-achieving women and men rank a STEM degree either first or anywhere in their application lists. In contrast to the variation we observe in the pipeline gap, the choice gap is large and consistently negative across all settings and definitions. Focusing on students in the top 10% and first-choice rankings, women are on average 23.5 percentage points less likely than similarly talented men to rank STEM first; the gap ranges from about 12 percentage points in Uganda to about 36 percentage points in China. Closing performance gaps is therefore unlikely to be sufficient without also addressing systematic differences in choice. Indeed, even in countries where women are well-represented among high-achieving students—such as Sweden, where they constitute a majority—they remain underrepresented in STEM applications due to the persistent choice gap. This observation underscores that reducing the pipeline gap, while important, is not enough to equalize STEM representation without also directly addressing the choice margin.

The stability of the choice gap across contexts as different as Chile, Greece, Sweden and Taiwan suggests that the underlying drivers are not context-specific, but reflect elements present across very diverse settings. Prior research highlights gendered preferences: women place greater weight on non-pecuniary attributes such as stability and flexibility ([Wiswall and Zafar, 2018](#)), and differences in program tastes explain much of the field gap ([Zafar, 2013](#)). Pipeline factors alone cannot account for persistent underrepresentation ([Patnaik et al., 2021](#)). Our findings extend this literature by showing that these differences are not context-specific, but persist across societies with widely varying levels of development and gender norms.

Understanding the mechanisms behind this stable choice gap remains a central challenge. It may reflect differences in how men and women value job

attributes, expectations of discrimination, family considerations, identity and belonging, or self-efficacy. Our results suggest that these forces operate broadly across settings rather than being tied to specific institutional contexts. Policies that address them directly could therefore play an important role in advancing gender parity in STEM—an objective with implications not only for equity but also for realizing the efficiency gains from a more inclusive allocation of talent.

Finally, a further contribution of this paper is to adopt a *market design perspective*, using administrative microdata from centralized admissions systems based on deferred acceptance (DA). Because these systems collect applicants’ full rank-ordered lists and truthful reporting is a dominant strategy under DA, they provide unusually good proxies for revealed preferences at the point of application. Restricting attention to high-achieving students proxies access to competitive STEM programs and isolates choice differences conditional on eligibility. At the same time, these preferences are themselves shaped over the life cycle—by families, schools, social norms, and labor-market expectations—so understanding when and how these differences emerge remains a key agenda for future work. As digital application platforms expand, the same approach can be used not only to study gender disparities, but also racial, socioeconomic, and other inequalities in higher education at scale.

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The Global Gender Gap in STEM

Applications: Pipeline vs. Choice

Online Appendix

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Appendix A. Institutional Details

This section provides additional details on the admission systems through which colleges select their students in the educational systems that we study. It also provides information on the data sources we use in this project.

Appendix A.1. Australia

Australian states use separate clearinghouses that operate using similar rules. Our study focuses on the clearinghouse in the state of Victoria. Each student submits a ranked ordered list of up to 12 college-major combinations to the clearinghouse, which allocate them to the highest-ranked combination to which they are eligible based on the student rank in the applicants' cohort (an "Australian Tertiary Admissions Rank" (ATAR)). ATAR uses a combination of scores for a variety of subjects with a possible small adjustment based on the student's affirmative action status. As two students may have zero subjects in common, we use ATAR as the measure of academic performance. A high school subject result combines an end-of-academic-year exam, administered in November, and the results of the tests throughout the last year of high school. Students finalize their ranked list once they receive their ATAR in December, are offered admission in January, and have a few weeks to enroll.

Our data for Australia comes from the Victorian Tertiary Admissions Centre, a clearinghouse that processes the applications. It includes individual-level information on all students applying to tertiary institutions in 2009 and 2010. The variables include performance in English and math subjects, applicants' ranked choices, gender, and parental education, which we use to proxy socioeconomic status. We classify students whose parents at most completed high school as low-SES, and those who have a parent with a bachelor's degree as high-SES.

Appendix A.2. Brazil

All public tertiary institutions in Brazil can opt to participate in a centralized digital platform that matches students to degree programs (SISU).⁶ The students are assigned to degree programs using an iterative deferred acceptance algorithm that takes into account performance in the national university entrance exam (ENEM), student preferences, and affirmative action status.⁷ The

⁶Centralized matching for public universities was introduced in 2010. By 2016 - the year of our data - 57% of the 4.8 million university entrance exam takers applied to a degree program using the SISU platform.

⁷To implement affirmative action, each degree sets aside quotas reserved for students from public high schools. As a result, targeted students face lower admission cutoffs than non-targeted students.

ENEM takes place at the end of the Brazilian academic year in December. Students submit their preferences in the first week of January, receive their results the third week of January and have until the last week of January to accept or reject an offer. Students can submit up to two choices in their application form. Private universities use decentralized admissions right after the assignment by SISU.

Our data for Brazil cover the year 2016 and are made up of two datasets in which we observe all applications submitted to the centralized admission system and students’ test scores. The data include students’ overall and subject-level scores, students’ ranked preferences for specific college-major combinations, demographic characteristics—i.e., gender and age—and self-reported household income.

Appendix A.3. Chile

Chile uses a nationwide centralized admissions system that in the period we study covered all public universities and 17 out of the 43 private universities.⁸ As in the case of Brazil, students are allocated to specific college-major combinations through a deferred acceptance admission algorithm. Students submit a list of up to 10 preferences and are then allocated the highest one for which they are eligible based on their high school GPA, their performance in a national level college admission exam (PSU), and affirmative action status.⁹ In contrast to Brazil, less than 4% of the seats offered by the universities that participate in centralized admissions are reserved for low-SES students. Students take the PSU in early December, at the end of the Chilean academic year, but register for the examination in mid-August. Students receive their scores and apply for college using an online platform by the end of December. In early January they learn where they were admitted and have a couple of weeks to decide whether to take or reject the offer.

Our data for Chile comes from the Chilean agency in charge of college admissions, DEMRE, and includes individual-level information on all students who registered to take the PSU between 2004 and 2018. The data include students’ performance in high school and on each section of the PSU; students’ ranked

⁸In total there are 60 universities in Chile. Before 2012 only 9 private universities participated in the centralized admission system. In 2012, eight new institutions joined the system. Since 2020, most universities use the centralized admission system. Admissions to two-year colleges are fully decentralized.

⁹In 2018—the last year we observe for Chile—the registration fee for the PSU was around USD 47. As of 2006, all public and voucher school graduates (93% of high school students) are eligible for a fee waiver that makes the PSU free for them. The entire registration process operates through an online platform that automatically detects the students’ eligibility for the fee waiver.

application choices; demographic characteristics (gender and age); and parental education, which we use as a proxy for socioeconomic status. We classify students whose parents at most completed high school as low-ses, and students whose parents had some university education as high-ses.

Appendix A.4. China (Ningxia Hui Autonomous Region)

China, including the Ningxia province, uses a college admissions system centralized at the province-track level. Students only compete for college-major vacancies with peers within the same province. All Chinese provinces match students to tertiary education places using the Chinese Parallel mechanism, which is similar to deferred acceptance, and takes into account students' choices and their academic performance in a national entrance exam (College Entrance Examination, CEE).¹⁰ The process begins with the administration of the CEE in early June. In mid-June, all Chinese colleges publish their college-major allocation quotas for each province and their tuition fees. After learning their CEE score and the college-major quotas and fees, students submit their applications with up to 90 ranked choices at the end of June.¹¹ Students who choose not to pursue tertiary education do not submit applications.

Following the matching process, each student receives a single take-it-or-leave-it offer. If a student declines the offer or does not get any offers, they must wait until the following year to retake the CEE. The alternative is to enter the labor market with only a high school degree or to enroll at a tertiary institution abroad.¹²

We use data for the province of Ningxia for the year 2018, which is supplied by the Ningxia Department of Education. The data includes students' performance on the CEE; their ranked application choices; demographic characteristics (gender and age); and parental education that we use as a proxy for socio-economic status. As in the case of Chile, we classify students whose parents at most completed high school as low-SES, and students whose parents had some university education as high-SES. Ningxia is among the poorest provinces in China, yet application choices of students remain representative of those seen in the rest of the country (see for example [Ding et al., 2021](#), [Loyalka et al., 2021](#)).

¹⁰See discussion in [Chen and Kesten, 2017](#) on the Chinese Parallel mechanism.

¹¹In 2018, 42% of students from Ningxia remained in Ningxia. This number ranges from 15%-19% in all provinces. Of all students attending tertiary education in Ningxia, 35% are from other provinces. This number ranges from 10-80% in all provinces.

¹²Few students choose to go overseas after having taken the CEE. Those who aim to study abroad usually do not take the CEE and most of them have enrolled in foreign institutions before the CEE in June.

Appendix A.5. Finland

Most tertiary education institutions in Finland participate in a centralized college admissions system.¹³ To assign students to seats, the system takes into account students' preferences, their high school exit examination, and/or a university entrance examination.¹⁴ The assignment algorithm goes through queues in a pre-specified order and students are allocated to places by applying a deferred acceptance algorithm. The main application round takes place in spring. Finnish students submit their applications with their ranked choices a few months before sitting their university entrance exam and before knowing their final high school GPA.¹⁵ Individuals submit applications through an online platform that is open for around two weeks in March. Students can rank up to six college-major combinations in their application.¹⁶ The online platform also contains information on how different college-major combinations weight high school exit exams, program-specific queues, and whether there exist any special requirements. Once an offer is made, students can decide whether to accept the offer and enroll in that college-major or not.

Our data for Finland come from two registers maintained by the Ministry of Education—HAREK and AMKOREK—for the years 2016 to 2020. In these data, we observe the full set of applications, whether a student took the college entrance examination, high school GPA, and high school exit examination grades. We augment this dataset with information from Statistics Finland on the student's age and parental education. As in the case of Chile and China, we classify students whose parents' highest level of education is high school as low-SES. Students with at least one parent completing university and a master's degree are classified as high-SES.

Appendix A.6. Greece

Most tertiary education institutions in Greece are public and use a centralized admissions system.¹⁷ Students are assigned to tertiary education places

¹³36 out of the total 38 tertiary education institutions are part of the centralized admissions system.

¹⁴Before 2020, most of the programs admitted students through three queues: the first queue considered only exit exams, the second queue ranked individuals based on a joint score of high school exit and university admission exams, and finally, some individuals were admitted only based on the university admission exams. In 2020, the joint score queue was abolished, and currently, individuals are admitted based on their high school exit exams or university admission exams only.

¹⁵Applications take place twice a year, once in the fall and once in spring, but in this paper, we focus on the spring period since the number of programs offered during the fall round is very limited.

¹⁶In earlier years, students were able to list more courses.

¹⁷There are a number of private universities using a fully decentralized admissions system. However, graduates of private institutions do not yet have equal degree recognition rights as

using a deferred acceptance admissions algorithm that takes into account performance in a college admissions examination (Panhellenic Examinations, PE), student preferences, and affirmative action status.¹⁸ Well in advance of the PE, the Ministry of Education publishes the number of places for each college-major and the subject weights assigned to different parts of the PE.¹⁹ Students sit the Panhellenic Examinations in May-June and after getting their results submit their application forms with their ranked preferences in early August. There is no limit to the number of choices. However, students can only apply to at most 2 out of the 5 general subject categories into which college majors are divided. In practice, this translates to about 350 choices out of 600 available.

Each student receives a take-it-or-leave-it offer and if a student declines the offer, they can retake the exam the following year at no financial cost.²⁰ Some students do not receive any offers and can also retake the following year. In our analyses, we focus on first-time applicants.

Our data for Greece are supplied and maintained by the Ministry of Education and include the universe of applicants. For each student, we observe their full set of choice lists, their final PE score, their gender, and the high school they attended for the years 2003-2012. From the Ministry of Finance, we obtained a measure of average household income at the postcode level for the year 2009, which we match to the location of students' high school. We use the latter information as a proxy for socioeconomic background. We define low-SES students as those coming from the bottom third of the income distribution and high-SES those coming from the top third.

Appendix A.7. Spain

Spain has 86 universities, 50 public and 36 private. Private universities have a fully decentralized admission system, while public universities select their students using a centralized deferred acceptance admission system that takes into account a weighted average of the high-school GPA and a college entrance exam (EBAU).²¹ Most students enroll in a public university program (83%). The

graduates of public institutions within Greece.

¹⁸Affirmative action works by reserving a pre-specified proportion of college places for specific sub-groups, and affects around 5% of applicants. Thus students from a specific sub-group that benefits from affirmative action compete only with other students with the same status.

¹⁹The PE consists of 6-9 papers that are centrally set and graded.

²⁰Students can also retain their first PE score and re-use the following year. However, they will not be competing with the main cohort, only with other students who also chose to defer their entrance.

²¹The final grade for each student is computed as a weighted average of two-year high school GPA (60% of the total score) and the entrance exam (40% of total score). Students can gain extra points by taking elective subjects in the entrance exam. The final score ranges between 0 and 14 points, where the threshold of passing is established at 5 points.

students take the entrance exam at the beginning of June, which corresponds to the end of the Spanish academic year. Students apply to their majors of interest using an online platform open between mid-June and the beginning of July and can submit a rank-ordered list of up to 12 college-majors combinations. Colleges publish the list of majors and vacancies offered for the next academic year before the application starts. The results of the first round of admissions are published in mid-July. Students can either accept their initial offer and pre-register or reject it. If they reject this initial offer, they could end with no offer or with an offer for an option far down in their priority list.²²

Our data for Spain cover students who take the EBAU to access university (81% of an average cohort) between 2018 and 2020 and come from the Spanish Ministry of Universities.²³ We observe students' GPA in high school, their score in the EBAU, and their list of preferences. We also observe their age and gender. As in other countries, we approximate socioeconomic status with parental education: students whose parents at most completed high school are classified as low-SES, while students whose parents completed a higher education degree are classified as high-SES.

Appendix A.8. Sweden

In Sweden, postsecondary education is tuition-free and all students are eligible for a monthly stipend as well as a subsidized loan. Admission to tertiary education in Sweden is centrally managed by Universitets och Högskolorådet (UHR). Students are matched to tertiary education places using a deferred acceptance algorithm that takes into account students' ranked preferences, their high school GPA, and/or their university entrance examination (Högskoleprovet). As in Finland, but unlike the college admission exams of the other countries in the study, the university entrance exam is voluntary. The system contemplates multiple *admissions queues* and students can participate in all of them simultaneously. For each program, at least a third of the vacancies are reserved for a group that only competes based on high school GPA. At least another third is allocated based only on the university admission exam. The remaining third of vacancies are also mostly assigned by high school GPA, but can sometimes be used for

²²If a student rejects the offer, she can apply to a different program. If the program is over-subscribed the student is placed on the waiting list even if she has a higher score than others currently admitted because she modified her order of preference after the deadline of the first applications.

²³Students not entering university through the standardized entrance exam represent 19% of first-year students. 53% of them come from the vocational track of tertiary education. The other 47% of them already completed a university degree, come from foreign universities, or took special exams designed for students older than 25, 40, and 45 years old.

special admission paths.²⁴

Applicants' best ranking determines their admission status. Two rounds of applications are organized each year, a larger one in April for programs and courses starting in August and one in October for those starting in January.²⁵ As in Finland, most students submit their rank of preferences before knowing their high school GPA. Students may rank up to 20 alternatives in each application round. After an initial round of allocations, applicants can choose to accept any offer they receive or participate in the second round for admission to higher-ranked alternatives.²⁶ If a first-round offer is declined, it cannot be recovered.

The data for Sweden come from the Swedish Council for Higher Education (UHR) and cover the years 2008 to 2017. The data contain information on students' high school GPA, their scores on the college admission exam, and their rank of applications. We match individual records to data from Statistics Sweden to obtain information on their gender, age, and parental education. As in previous cases, we classify students whose parents' highest level of education is high school as low-SES and students whose parents attended university as high-SES.

Appendix A.9. Taiwan

In Taiwan, all the public and private universities participate in a centralized clearinghouse operated by the Joint Board of College Recruitment Commission (JBCRC). The admission unit is a university-major combination (termed as a "program"). Students need to take two exams, General Scholastic Ability Test (GSAT) in January/February and Advanced Subjects Test (AST) in July. After knowing their scores, students submit their rank-ordered lists (ROLs) to the clearinghouse. The maximum number of choices that each student can rank is 100 in the most recent academic year. University programs rank students by evaluating students' exam performance mainly in AST. Students are assigned to programs via a program-proposing Deferred Acceptance (DA) algorithm. The final placement results are announced in August. Students can only enroll in the program to which they are assigned (or opt out of the system) as there is full compliance strictly executed (with very few exceptions applied to top-scoring students).

²⁴This is the case in some highly selective majors, where an additional test or an interview is sometimes used to allocate this last third of vacancies. We do not include admissions through such groups in our analysis.

²⁵Students can apply to full-time programs and short courses in the same application. A student can never be admitted to multiple programs in the same semester, but could be admitted to both a course and a program, or multiple courses.

²⁶Their scores and lists of preferences do not change between the two rounds, but the admission cutoffs might.

Our data for Taiwan come from JBCRC’s administrative registers and cover years 1996-2003. We observe students’ exam scores of all the subjects in both GSAT and AST, their reported ROLs, and final admission outcomes. Augmented with various insurance records from Taiwan’s Ministry of Labor, we also observe rich information on students’ demographics and family background, such as age, gender, residential location, parental income and education level. Furthermore, attributes of high schools from which students graduated are supplied by the Ministry of Education. We define a student to be low-SES if her parents at most completed high school, and high-SES if her parents had some university education.

Appendix A.10. Uganda

As of 2018, Uganda had eight public universities and 44 private universities. According to the National Commission on Higher Education, more than half of university enrollments are in public universities. Students can be admitted to public universities through two schemes: the national merit scholarship, which covers both living expenses and tuition, and the self-funding scheme, where students pay for both tuition and living expenses. The national scholarship scheme is centralized, while the self-funding scheme and admissions to private universities are decentralized. Admission to universities under either scheme is based on test scores, specifically the "weighted score" of national exams, which are similar to subject-based SATs in the U.S.²⁷ Students apply to the centralized scheme before knowing their test scores, typically between December and January, with some even applying before taking their national exams in early December. Offers are usually announced around March or April. Unlike in many other countries, students in Uganda take exams in three subjects, known locally as the "subject combination," when leaving secondary school. This combination may vary by student. For example, one student might take exams in biology, chemistry, and math, while another might take physics, economics, and math exams. Thus, students’ tracks (STEM or non-STEM) are determined early during upper secondary school. A student who takes a non-STEM subject combination cannot apply for STEM majors at university, but the reverse is possible—a student with a STEM subject combination can apply to both STEM and non-STEM programs. However, before selecting their “A-level” secondary subject combina-

²⁷There are additional schemes of admissions, such as district quota, disabled quota, sports, international students, mature age, and diploma or degree holders. Still these categories depend on test in national exams and make up a very small portion of the total admissions and are mostly self-funded, except for the district and disabled quotas. Also, more recently, the largest public university has implemented a gender affirmative action policy that sets quotas for either gender to at most 60% of the slots in each major.

tion, students must take national exams in at least eight subjects at the end of “O-level” (similar to junior high school). These exams can qualify students for specific subject combinations and influence their future career paths in STEM, social sciences, humanities, or business. Our data from Uganda covers university applicants through the centralized platform, the Public University Joint Admissions Board (PUJAB), spanning at least seven years. This dataset includes each student’s preference list, ranking six majors in order of preference, along with their test scores in the subject combinations and scores in the O-level national exams. We also have information on each applicant’s age, gender, high school, and district of origin. Most importantly, we observe universal test scores from O-level school exams, which allow us to determine the cut-off scores for the students in top 10% of the score distribution. We then use this cutoff to observe if applicant scores in O-level English and Math were in the top 10% of the score distribution.

Appendix B. Pipeline and STEM choice gaps along the academic performance distribution

This section extends the analysis presented in the main body of the paper to include as well students below the top 20% of the academic performance distribution. Our focus on top students in the main text is motivated by the fact that these students are most likely to gain admission to and succeed in selective STEM programs—i.e., the ones associated with high earnings. In addition, these students are the less affected by strategic concerns in settings that limit the number of programs students can list. However, it is also informative to examine whether the patterns we document for top students are also present in the rest of the distribution.

Appendix B.1. Pipeline gap across score bins

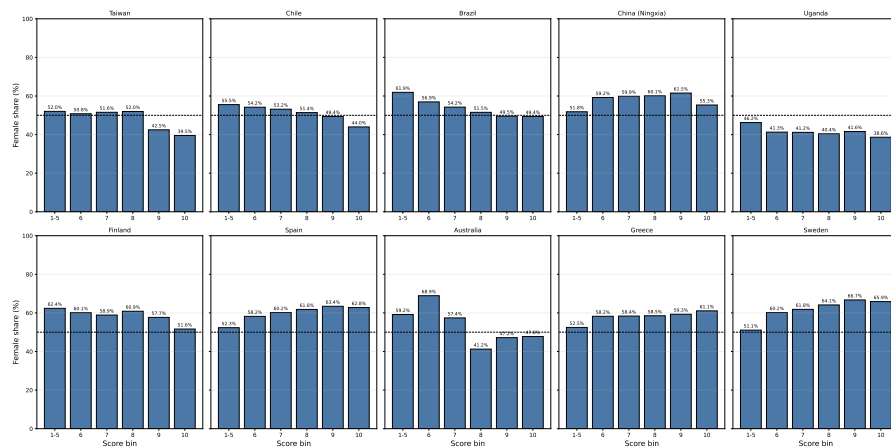
In this appendix, we extend the analysis from top-performing students to the broader score distribution while preserving more detail in the upper tail, where admission to selective STEM programs is most plausible. Specifically, we pool deciles 1–5 into a single “bottom 50%” group and keep deciles 6–10 separately. This reduces noise in the lower half of the distribution, where students are less likely to qualify for selective STEM tracks, while maintaining a decile-by-decile decomposition among higher-performing applicants.

Figure B.I reports female shares across these bins. The pattern varies across countries. In some settings—for example, Brazil, Chile, and Finland—female representation tends to decline with academic performance, while in others—for example, Greece and Sweden—it tends to increase. This heterogeneity is consistent with the main-text evidence that pipeline differences vary by context.

Appendix B.2. Choice gaps across score bins

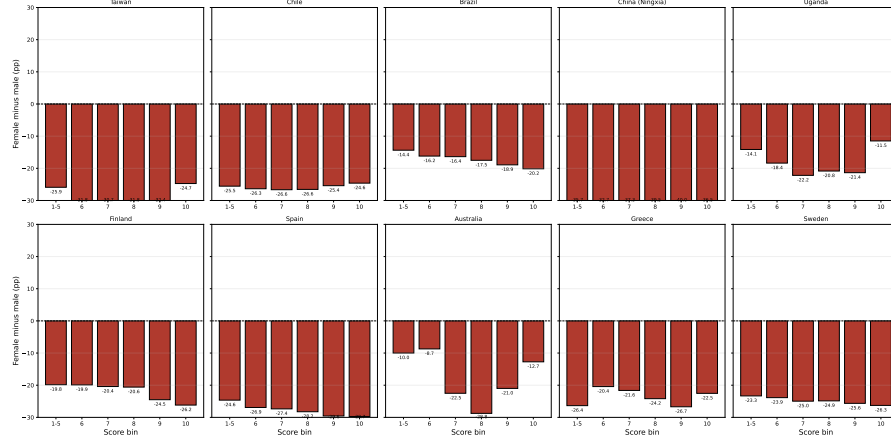
Figures B.II and B.III report the female-minus-male gap in STEM application probabilities by score bin, using first-choice and any-choice definitions, respectively. In both figures, the gap is negative in virtually all countries and bins. The magnitude of the gap varies with performance in some settings, but the sign and overall direction are remarkably stable. This reinforces the interpretation in the main text. The cross-country regularity is not that pipeline differences look similar, but that conditional STEM choice differences are persistently negative for women.

Figure B.I: Female shares in the bottom 50% and deciles 6–10 of the academic performance distribution



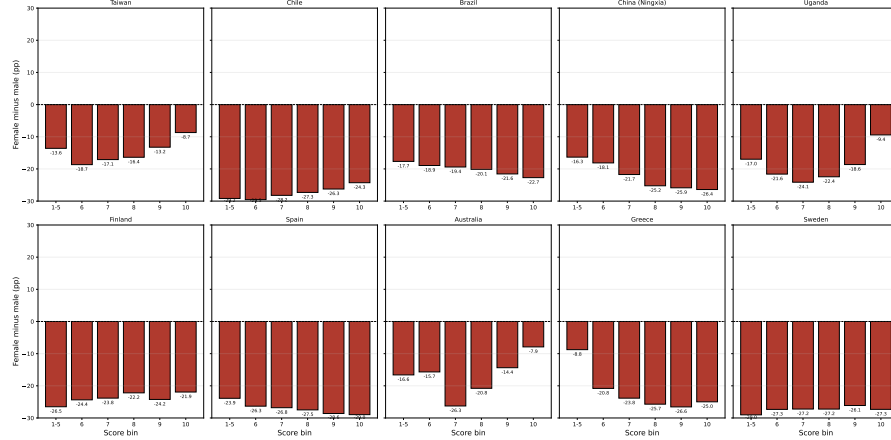
Notes: Each panel reports the female share (in percent) among applicants by score bin. The first bar pools deciles 1–5 (bottom 50%), and the remaining bars correspond to deciles 6, 7, 8, 9, and 10 separately. Deciles are computed within each country sample. The dashed horizontal line marks 50% female representation. Country samples follow the preferred harmonized variants used in the main analysis: Australia (reply_docx), Brazil (reply_tables), Chile (chile_2012_2018), China (china_ningxia_2018), Finland (main_pooled), Greece (years_2003_2012), Spain (esp_2020_2024_tables), Sweden (reply_docx_2017), Taiwan (reply_pdf_2010), and Uganda (reply_docx). Total sample sizes (all deciles, female + male) are: Australia 77,648; Brazil 2,711,049; Chile 1,747,925; China 52,427; Finland 347,904; Greece 680,035; Spain 857,870; Sweden 108,237; Taiwan 72,067; Uganda 36,247.

Figure B.II: STEM first-choice gap in the bottom 50% and deciles 6–10



Notes: Bars report the female-minus-male gap (percentage points) in the probability of ranking a STEM degree first, by score bin and country. The first bar pools deciles 1–5 (bottom 50%), and the remaining bars correspond to deciles 6, 7, 8, 9, and 10 separately. Probabilities are computed using sex-specific bin denominators from the pipeline (total applicants) and numerators from STEM first-choice counts. Negative values indicate lower first-choice STEM application rates among women. The same preferred harmonized country variants listed in Figure B.I are used here. Total sample sizes (all deciles, female + male) are: Australia 77,648; Brazil 2,711,049; Chile 1,747,925; China 52,427; Finland 347,904; Greece 680,035; Spain 857,870; Sweden 108,237; Taiwan 72,067; Uganda 36,247.

Figure B.III: Any-STEM gap in the bottom 50% and deciles 6–10



Notes: Bars report the female-minus-male gap (percentage points) in the probability of listing at least one STEM degree anywhere in the ranked application list, by score bin and country. The first bar pools deciles 1–5 (bottom 50%), and the remaining bars correspond to deciles 6, 7, 8, 9, and 10 separately. Probabilities are computed using sex-specific bin denominators from the pipeline and numerators from STEM-any counts. Negative values indicate lower any-choice STEM application rates among women. The same preferred harmonized country variants listed in Figure B.I are used here. Total sample sizes (all deciles, female + male) are: Australia 77,648; Brazil 2,711,049; Chile 1,747,925; China 52,427; Finland 347,904; Greece 680,035; Spain 857,870; Sweden 108,237; Taiwan 72,067; Uganda 36,247.

Appendix C. The choice gap in other selective degrees

This section presents the same choice-gap exercises as in the main body of the paper, but for other selective fields that are also associated with high labor-market returns, namely business-law and health. We report results for students in the top 5%, top 10%, and top 20% of each country’s academic performance distribution, and for both first-choice and any-choice outcomes.

Appendix C.1. Business and law

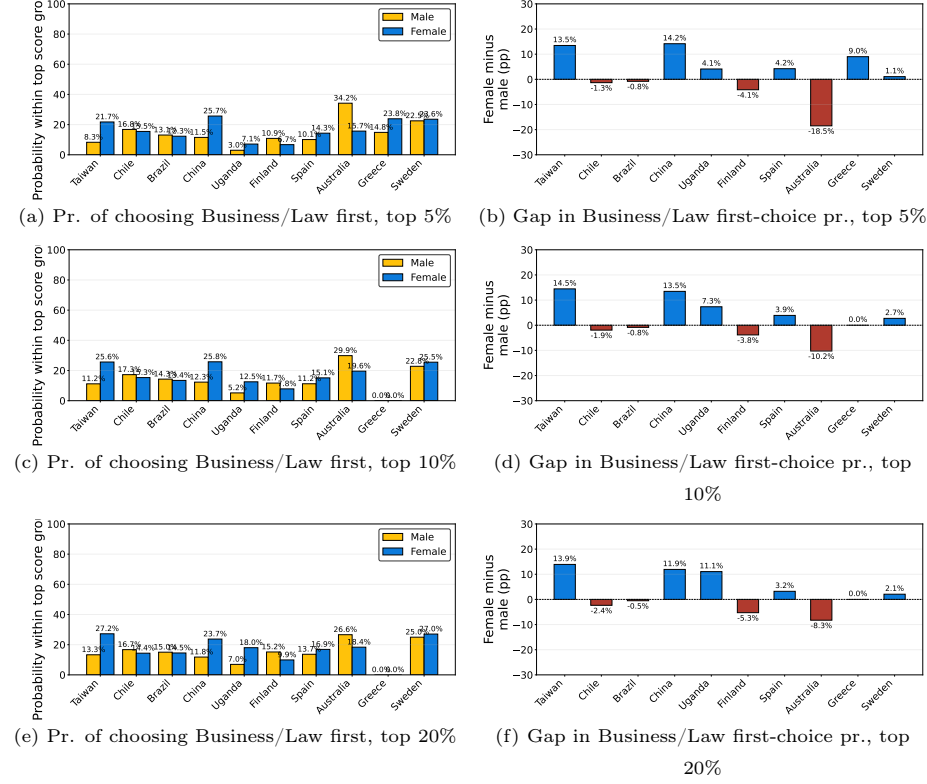
Figures C.I and C.II report female-minus-male gaps in business-law for first-choice and any-choice outcomes. In contrast with STEM, business-law does not exhibit a stable directional pattern across countries or score thresholds. In some settings the gap is positive, indicating that high-achieving women are more likely than comparable men to include business-law options in their rank-order lists; in other settings the gap is negative, indicating the opposite. The magnitudes are also heterogeneous, but they are typically smaller than the corresponding STEM choice gaps documented in the main text. Taken together, the business-law evidence is consistent with field-specific heterogeneity rather than a uniform gender sorting mechanism among selective programs.

Appendix C.2. Health

Figures C.III and C.IV report female-minus-male gaps in Health for first-choice and any-choice outcomes. Health displays variation across settings, but the central pattern is qualitatively different from STEM. In six out of ten country settings, we observe a positive choice gap, meaning that high-achieving women are more likely than similarly talented men to rank a health degree. In the remaining four settings, estimated differences are small and do not indicate a large male advantage. Thus, the health pattern is best characterized as either near parity or a female-tilted gap.

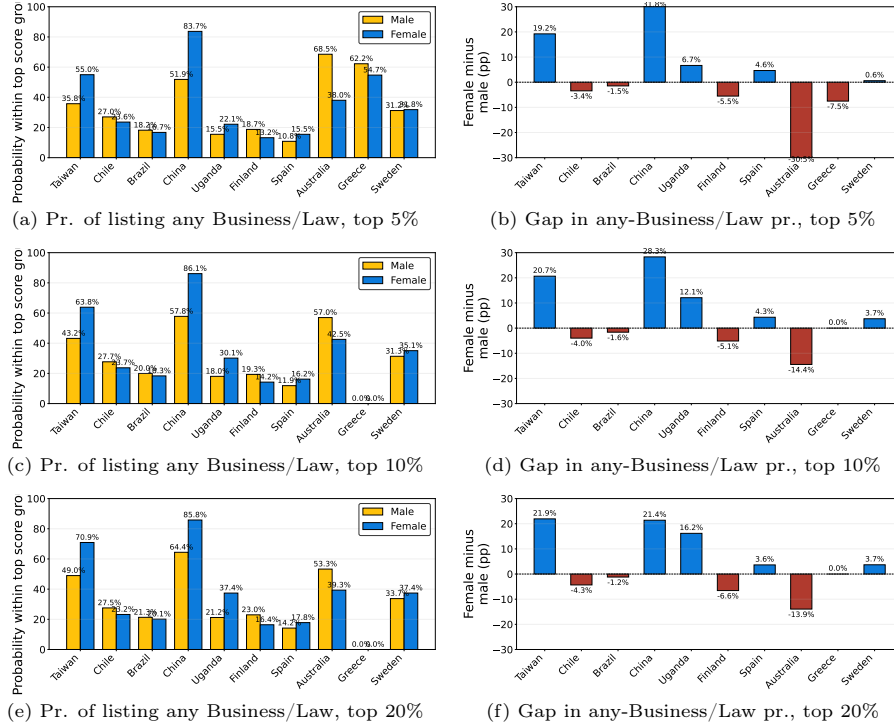
Taken together, the evidence in sections C.1 and C.2 indicates that the STEM gap is field-specific. The strong and systematic STEM disadvantage for women does not generalize to other high-return selective fields: business-law is mixed in sign and usually smaller in magnitude, while health is either close to zero or positive for women in most settings.

Figure C.I: Business/Law first-choice probabilities and gender gap across the academic performance distribution



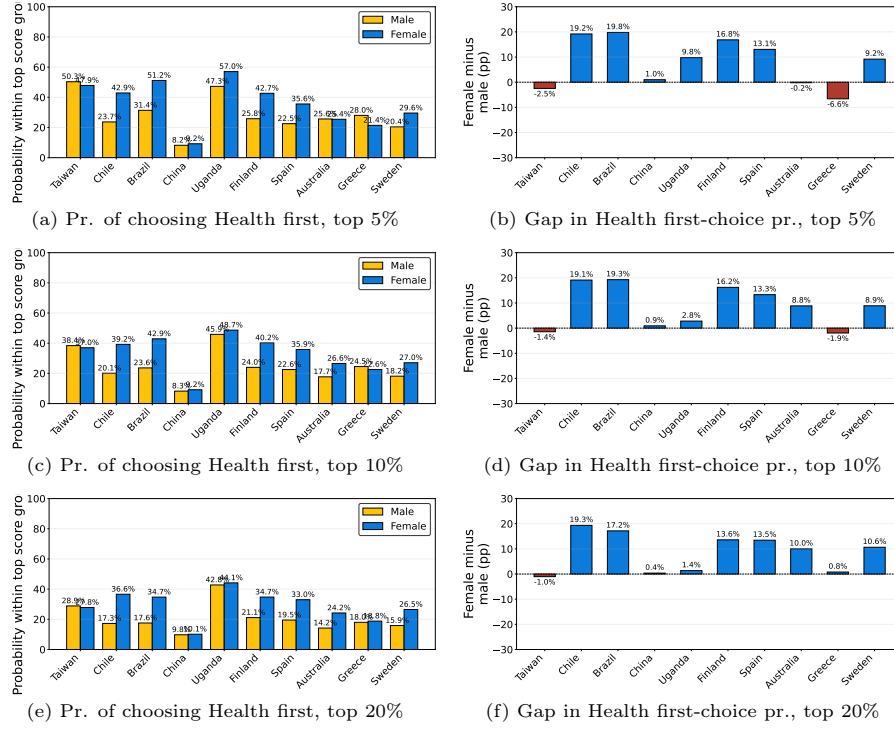
Notes: Panels (a), (c), and (e) show, by country, the probability that applicants rank a Business/Law program first in their preference list, separately for male and female students within the top 5%, top 10%, and top 20% of the academic performance distribution. Panels (b), (d), and (f) show the corresponding gender gap defined as female probability minus male probability (percentage points). Negative values indicate lower Business/Law first-choice propensity among women. Percentage labels on bars report exact values; in gap panels labels are above positive bars and below negative bars. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan (N=72,067), Chile (N=1,747,925), Brazil (N=2,711,049), China (Ningxia) (N=52,427), Uganda (N=36,247), Finland (N=347,904), Spain (N=857,870), Australia (N=77,648), Greece (N=680,035), and Sweden (N=108,237). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

Figure C.II: Any-Business/Law application probabilities and gender gap across the academic performance distribution



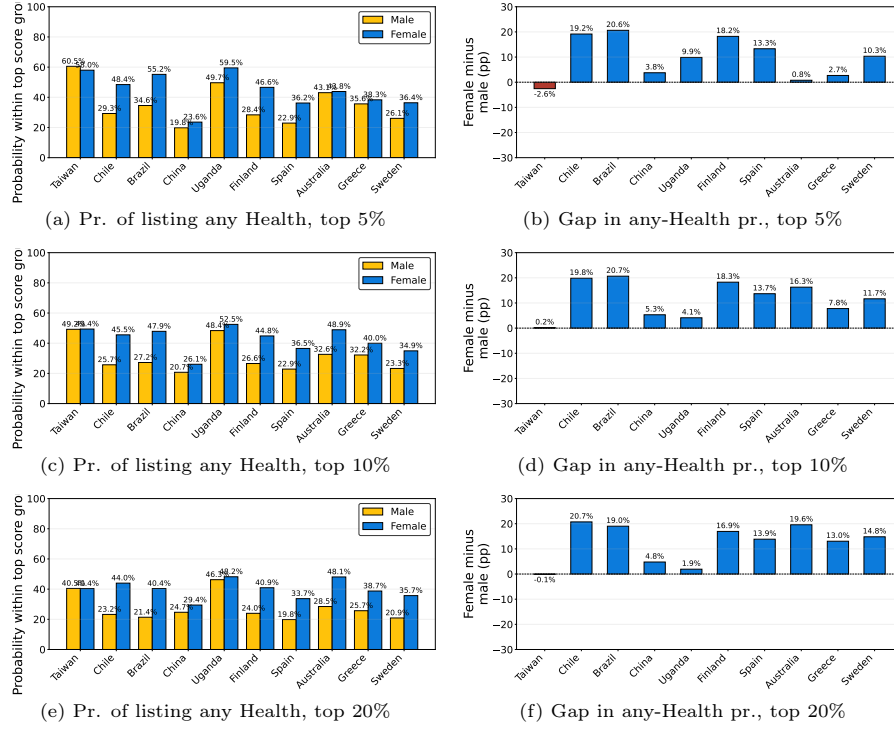
Notes: Panels (a), (c), and (e) report, for each country, the probability that an applicant includes at least one Business/Law program anywhere in the rank-order preference list, by sex and performance group (top 5%, top 10%, and top 20%). Panels (b), (d), and (f) report the female minus male difference in that probability (percentage points). Values are normalized by the number of applicants of each sex in the corresponding top-performance group. Percentage labels on bars report exact values; in gap panels labels are above positive bars and below negative bars. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan ($N=72,067$), Chile ($N=1,747,925$), Brazil ($N=2,711,049$), China (Ningxia) ($N=52,427$), Uganda ($N=36,247$), Finland ($N=347,904$), Spain ($N=857,870$), Australia ($N=77,648$), Greece ($N=680,035$), and Sweden ($N=108,237$). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

Figure C.III: Health first-choice probabilities and gender gap across the academic performance distribution



Notes: Panels (a), (c), and (e) show, by country, the probability that applicants rank a Health program first in their preference list, separately for male and female students within the top 5%, top 10%, and top 20% of the academic performance distribution. Panels (b), (d), and (f) show the corresponding gender gap defined as female probability minus male probability (percentage points). Negative values indicate lower Health first-choice propensity among women. Percentage labels on bars report exact values; in gap panels labels are above positive bars and below negative bars. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan (N=72,067), Chile (N=1,747,925), Brazil (N=2,711,049), China (Ningxia) (N=52,427), Uganda (N=36,247), Finland (N=347,904), Spain (N=857,870), Australia (N=77,648), Greece (N=680,035), and Sweden (N=108,237). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

Figure C.IV: Any-Health application probabilities and gender gap across the academic performance distribution



Notes: Panels (a), (c), and (e) report, for each country, the probability that an applicant includes at least one Health program anywhere in the rank-order preference list, by sex and performance group (top 5%, top 10%, and top 20%). Panels (b), (d), and (f) report the female minus male difference in that probability (percentage points). Values are normalized by the number of applicants of each sex in the corresponding top-performance group. Percentage labels on bars report exact values; in gap panels labels are above positive bars and below negative bars. Country-level sample sizes used in these figures (preferred sample variants, based on total pipeline counts across deciles 1–10) are: Taiwan ($N=72,067$), Chile ($N=1,747,925$), Brazil ($N=2,711,049$), China (Ningxia) ($N=52,427$), Uganda ($N=36,247$), Finland ($N=347,904$), Spain ($N=857,870$), Australia ($N=77,648$), Greece ($N=680,035$), and Sweden ($N=108,237$). Differences in sample sizes reflect both the size of each admissions system and the number of cohorts included in each country dataset.

Appendix D. Pipeline and STEM choice gaps in cohorts closest to 2018

Appendix D.1. Sample and time coverage

This appendix evaluates whether the main findings are sensitive to cohort composition by re-estimating top-10 pipeline and choice gaps using cohorts as close as possible to 2018 in each setting. The objective is to make the time period as comparable as possible across countries, although this is not feasible everywhere given differences in the underlying source files.

The country-year selections used are: Australia (2009–2010 pooled), Brazil (2016), Chile (2018), China (2018), Finland (2018), Greece (2003–2012 pooled), Spain (2020), Sweden (2017), Taiwan (2010), and Uganda (2021/2022).

Relative to the main-text samples, we move closer to 2018 in Chile (from 2012–2018 pooled to 2018), Finland (from 2016–2020 pooled to 2018), and Spain (from 2020–2024 pooled to 2020). In the remaining settings we use the same period as in the main text because no closer harmonized cohort is available in the current country files. In several cases this is still reasonably close to 2018: China is exactly 2018, Sweden is 2017, and Brazil is 2016.

Appendix D.2. Pipeline gap in recent cohorts

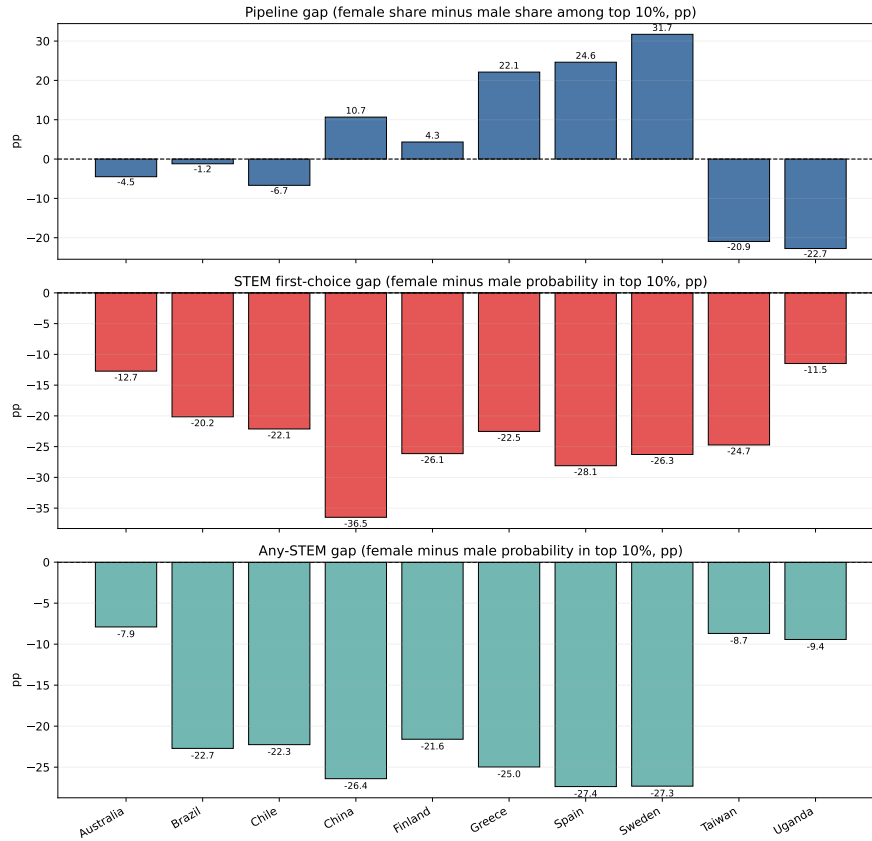
In Figure [D.I](#), Panel 1 shows the top-10 pipeline gap for cohorts closest to 2018. As in the main text, the pipeline gap continues to vary substantially across countries and changes sign across settings.

Appendix D.3. Choice gaps in recent cohorts

Figure [D.I](#), Panels 2 and 3, report first-choice and any-choice STEM gaps in recent cohorts. The first-choice STEM gap remains negative and sizable in all countries, and the any-choice STEM gap is also negative in all countries, with magnitudes generally close to the first-choice benchmark. Overall, restricting to cohorts closest to 2018 leaves the core empirical pattern unchanged.

Figure D.I: Top-10 pipeline and choice gaps in cohorts closest to 2018

Appendix D: Top-10 gaps using cohort closest to 2018 in each country



Notes: Bars report female-minus-male gaps in percentage points for students in the top 10%

of each country's academic performance distribution. Panel 1 shows the pipeline gap (female share minus male share among top-10 students). Panel 2 shows the first-choice STEM gap (female minus male probability of ranking a STEM degree first, conditional on being in the top-10 group). Panel 3 shows the any-choice STEM gap (female minus male probability of listing at least one STEM degree anywhere in the ranked application list, conditional on being in the top-10 group). Country-year selections are: Australia (2009–2010 pooled), Brazil (2016), Chile (2018), China (2018), Finland (2018), Greece (2003–2012 pooled), Spain (2020), Sweden (2017), Taiwan (2010), and Uganda (2021/2022). Top-10 sample sizes (female + male) are: Australia 7,668; Brazil 271,049; Chile 26,140; China 5,203; Finland 6,920; Greece 68,007; Spain 20,418; Sweden 10,758; Taiwan 6,998; Uganda 3,187.